

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Active-Comparator-Controlled Study of Bomedemstat Versus Hydroxyurea in Patients with Essential Thrombocythemia Naïve to Cytoreductive Therapy **Short Title/Acronym:** Shorespan-007 / MK-3543-007 **Protocol Number:** MK-3543-007 **NCT Number:** NCT06456346 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) / Merck & Co., Inc. **Investigational Product:** Bomedemstat (MK-3543 / IMG-7289) - LSD1 inhibitor **Phase of Development:** Phase III **Medical Monitor:** Gregory Lubiniecki, MD (Vice President, Global Clinical Development, Merck Research Laboratories)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of bomedemstat compared to hydroxyurea with respect to the durable clinicohematologic response (DCHR) rate in cytoreductive therapy-naïve patients with essential thrombocythemia (ET). **Secondary Objectives:** To evaluate the duration of hematologic remission, duration of clinicohematologic response, event-free survival, incidence of thrombotic/hemorrhagic events, disease progression rate (transformation to myelofibrosis [MF] or MDS/AML), and improvements in patient-reported outcomes including fatigue and total symptom scores.

2.2 Background & Rationale Essential thrombocythemia (ET) is a chronic, rare myeloproliferative neoplasm (MPN) often causing burdensome symptoms (fatigue, insomnia), thrombocytosis, and an increased risk of thrombosis or hemorrhage. The standard of care in ET has remained unchanged for decades, leaving an unmet need for novel therapies that not only improve disease control but also alter the natural history of the disease and improve patient quality of life. Bomedemstat is an orally available, irreversible lysine-specific demethylase 1 (LSD1) inhibitor that regulates the proliferation and maturation of hematopoietic stem and progenitor cells, demonstrating in phase 2 trials the ability to durably reduce platelet and white blood cell (WBC) counts and reduce mutation burdens.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Hydroxyurea) **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 300 evaluable patients **Number of Sites:** Global recruitment across sites in Asia, Australia, Europe, North America, and South America **Estimated Study Start:** 16-Jul-2024 **Estimated Primary Completion:** 28-Sep-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed diagnosis of Essential Thrombocythemia (ET) based on WHO Criteria for myeloproliferative neoplasms, with an indication for cytoreductive therapy. 3. Centrally assessed bone marrow fibrosis score of Grade 0 or Grade 1. 4. Platelet count $>450 \times 10^9/L$ and absolute neutrophil count $\geq 0.75 \times 10^9/L$. 5. Must be cytoreductive therapy-naïve (has received no prior cytoreductive treatment for their ET).

4.2 Exclusion Criteria 1. Documented increased risk of bleeding or a history of a bleeding diathesis. 2. History of any illness/impairment of gastrointestinal function that might interfere with oral drug absorption. 3. Active infection requiring systemic therapy. 4. History of a malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 5. Major surgery < 4 weeks prior to the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Experimental Arm:** Bomedemstat at a starting dose of 50 mg/day titrated to a target platelet count of $\geq 150 \times 10^9/L$ to $\leq 350 \times 10^9/L$ (plus hydroxyurea placebo). - **Active Comparator Arm:** Hydroxyurea at a starting dose of 500 mg/day titrated per the approved product labeling (plus bomedemstat placebo). **Route of Administration:** Oral **Duration of Treatment:** Up to 52 weeks of the core treatment phase. Patients who complete 52 weeks of treatment will be eligible to continue in the extended treatment phase.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and concomitant medications for baseline co-morbidities. **Prohibited:** Any prior or concurrent use of cytoreductive therapies (other than the assigned study interventions) for the treatment of ET.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, ET Diagnosis Confirmation, Medical History, Baseline Labs, Bone Marrow Assessment
Baseline	Day 0	Randomization, First	

Dose of Bomedemstat/Hydroxyurea, Safety Assessments | | **Interim Visits** | Weeks 2 to 12 (Every 2 weeks), then Every 4 weeks | Dose Titration, Safety Labs, Platelet & WBC Monitoring, Efficacy Assessment, PROMIS/MFSAF Patient-Reported Outcomes | | **End of Core Study** | Week 52 | Final Primary Endpoint Evaluation (DCHR), Extended Treatment Eligibility Review |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Durable Clinicohematologic Response (DCHR) rate. DCHR is defined as a confirmed reduction of platelet count to $\leq 400 \times 10^9/L$; absence of WBC count elevation to $>10 \times 10^9/L$ due to ET; absence of any thrombotic or major hemorrhagic events; and absence of disease progression to myelofibrosis (MF), myelodysplastic syndrome (MDS), or acute myeloid leukemia (AML) by Week 52. **Measurement Method:** Central and local laboratory hematology analysis, coupled with clinical event tracking.

7.2 Secondary & Exploratory Endpoints - Change in fatigue from baseline per the Myelofibrosis Symptom Assessment Form version 4.0 (MFSAF v4.0) individual fatigue symptom item score and PROMIS Fatigue SF-7a total score. - Change in total symptom score from baseline per the MFSAF v4.0. - Duration of clinicohematologic response and hematologic remission. - Event-free survival and incidence rates of thrombotic and major hemorrhagic events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs will be closely monitored throughout the study and for ≤ 30 days after treatment end. Safety events will be graded per the NCI Common Terminology Criteria for Adverse Events (CTCAE) v5.0. **Serious Adverse Events (SAEs):** Captured continuously and subject to standard rapid-reporting timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to review safety events and clinical outcomes routinely.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety and efficacy analyses will be conducted with all randomly assigned patients who receive at least 1 dose of study intervention. **Sample Size Justification:** Approximately 300 patients will provide adequate power to detect a statistically significant superiority of bomedemstat over hydroxyurea with respect to DCHR. **Handling of Missing Data:** The Stratified Miettinen-Nurminen method will be used to compare the endpoints of durable clinicohematologic response rate and incidence of disease progression between the two arms. Treatment

difference in event-free survival will be assessed using a stratified log-rank test.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring corresponding to the intense early visit schedule (every 2 weeks initially, then every 4 weeks). **Audit Trail:** Electronic data capture (eCRF) with stringent audit trails for adverse events, dose titrations, and central laboratory integration.

11. Study Identifiers & Governance

Primary ID: MK-3543-007 **Registry Number:** NCT06456346 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** Shorespan-007 **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Active-Comparator-Controlled Study of Bomedemstat Versus Hydroxyurea in Patients with Essential Thrombocythemia Naïve to Cytoreductive Therapy

12. Clinical Framework (Essential Thrombocythemia Specific)

Primary Condition: Essential Thrombocythemia (ET) **Diagnosis Threshold:** Platelet count $>450 \times 10^9/L$ **Medical Methodology:** Cytoreductive Therapy (LSD1 inhibition vs. Antimetabolite) **Optimization Target:** Normalization of Platelet count ($\geq 150 \times 10^9/L$ to $\leq 350 \times 10^9/L$) and WBC control without thrombotic/hemorrhagic events.

13. Study Procedures & Methodology

Management Pathway: Double-blind active comparator trial with individualized oral dose titration. **Education Protocol (HCP):** Site-level training on protocol-defined platelet target titrations and AE management. **Education Protocol (Patient):** Patient instruction regarding symptom tracking using PROMIS and MFSAF v4.0 e-diaries. **Quality Audit Mechanism:** Centralized laboratory review for bone marrow fibrosis grading and hematologic assessments to ensure rigorous outcome consistency.

14. Observation & Follow-Up Schedule

Total Duration: Up to 52 weeks (plus optional extended treatment phase) **Onsite Visit Cadence:** Every 2 weeks for the first 12 weeks, then every 4 weeks **Remote Monitoring Frequency:** Supplemental safety check-ins as defined by local site investigator protocols. **Checklist Review Interval:** Continuous monitoring of symptom assessments and laboratory changes every 4 weeks post-titration phase.

15. Sign-off & Approval

Role	Name	Signature	Date		---	---	---	---
Principal Investigator	[Insert Site PI Name]	_____	[DD- MMM-YYYY]					
Clinical Operations Lead	Gregory Lubiniecki, MD	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Statistician Name]	_____	[DD-MMM-YYYY]					